

Section 1: From Kantorovich to Wasserstein

The Wasserstein Distance

For a metric space $(\mathcal{X}, d)$, take $c(x, y) = d(x, y)^p, p \geq 1$. This yields the the **Wasserstein Distance**:

$$W_p(\alpha, \beta) = \left(\inf_{\pi \in \Pi(\alpha, \beta)} \int_X d(x, y)^p \, d\pi(x, y) \right)^{1/p}$$